

A Compound Statistical Model Based Radar HRRP Target Recognition*

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Abstract. In radar HRRP based statistical target recognition, one of the most challenging tasks is how to accurately describe HRRP's statistical characteristics. Based on the scattering center model, range resolution cells are classified, in accordance with the number of predominant scatterers in each cell, into three statistical types. In order to model echoes of different types of resolution cells as the corresponding distribution forms, this paper develops a compound statistical model comprising two distribution forms, i.e. Gamma distribution and Gaussian mixture distribution, for target HRRP. Determination of the type of a resolution cell is achieved by using the rival penalized competitive learning (RPCL) algorithm. In the recognition experiments based on measured data, the proposed compound model not only has better recognition performance but also is more robust to noises than the two existing statistical models, i.e. Gaussian model and Gamma model.

1 Introduction

A high-resolution range profile (HRRP) is the amplitude of the coherent summations of the complex time returns from target scatterers in each range resolution cell, which contains the target structure signatures, such as target size, scatterer distribution, etc., thereby radar HRRP target recognition has received intensive attention from the radar automatic target recognition (RATR) community [1],[2]. However, there has been little work in the field of radar HRRP based statistical recognition, in which how to accurately describe HRRP's statistical property is a challenging task.

Based on the scattering center model [3], range resolution cells are classified, in accordance with the number of predominant scatterers in each cell, into three statistical types. Therefore, in order to model echoes of different types of resolution cells as the corresponding distribution forms, this paper develops a compound statistical model comprising two distribution forms, i.e. Gamma distribution and Gaussian mixture distribution. Determination of the type of a resolution cell is achieved by using the rival penalized competitive learning (RPCL) algorithm. Experimental results for measured data show that the proposed model not only has better recognition perform-

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ance but also is more robust to noises than the two existing statistical models, i.e. Gaussian model [1] and Gamma model [2].

2 Statistical Models for Target HRRP

2.1 Statistical Characteristics of Target HRRP

According to the scattering center model [3], the limitation of target-aspect change to avoid scatterers' motion through resolution cells (MTRC) is given by

$$\delta\phi \leq (\delta\phi)_{\text{MTRC}} = \frac{\Delta R}{L} . \quad (1)$$

where ΔR is the length of range resolution cell, and L is the cross length of target. The immovable scatterers and their distribution in a range resolution cell are irrelevant to those in another cell within $(\delta\phi)_{\text{MTRC}}$, thus the resolution cells of an HRRP approximate to be statistically independent. Usually, there are three kinds of scatterer distribution in a range resolution cell.

a) First type of range resolution cell: There are a large number of small scatterers and no predominant scatterers in this type of range resolution cell. Under the hypothesis that the intensities of the small scatterers are almost same and the number of the small scatterers is large enough, the central limit theorem holds. If the echo phases of the small scatterers are uniform-distributed, according to the central limit theorem, the I- or Q-component of the complex echo of this type of cell will follow Gaussian distribution with zero mean and its amplitude will follow Rayleigh distribution.

b) Second type of range resolution cell: This type of range resolution cell consists of a predominant scatterer and a large number of small scatterers. Under the hypothesis similar to the first type, the I- or Q-component of the complex echo of this type of cell still follows Gaussian distribution with the mean determined by I- or Q-component of the echo of the predominant scatterer, in accordance with the central limit theorem. Thus its amplitude follows a Ricean distribution amplitude.

c) Third type of range resolution cell: There are a large number of small scatterers and several predominant scatterers in this type of range resolution cell. If a resolution cell has two predominant scatterers with their echoes' phase difference uniform-distributed, the amplitude will follow a distribution with double peaks whose values will be blunt by the echoes of the small scatterers. The echo of the third type of resolution cell usually follows a distribution with several peaks (mainly double peaks). However, relevant literatures [1],[2] didn't take such distribution into consideration. This paper will introduce the distribution with several peaks into HRRP based statistical recognition.

2.2 Statistical Description of Target HRRP

Now there exist two statistical models for target HRRP, i.e. Gaussian model [1] and Gamma model [2]. Gaussian model is based on that the power transformed echo per

resolution cell approximates to be Gaussian-distributed [6]. Two problems arise when Gaussian model is used. Firstly, the power parameter used in power transformation needs to be carefully selected in the training phase, otherwise, the recognition performance will be degraded seriously; Secondly, power transformed HRRPs are more sensitive to the signal-to-noise-ratio (SNR) than original HRRPs. Gamma model is motivated by the fact that Gamma distribution has been empirically observed to model target cross-section fluctuations under the low-resolution condition [4]. However, the statistical characteristics of target HRRP are much more complex than those of point-target's echo with low resolution radar. Statistical description of the three types of echoes will be discussed in the following.

a) Under the ideal condition, the echoes of the first and second types of resolution cells are Rayleigh-distributed and Ricean-distributed, respectively. What the ideal condition requires is a large number of the scatterers with almost same intensities per resolution cell, which ensures that the central limit theorem holds. However, increasing the resolution implies a reduction of the number of scatterers per resolution cell, thus the Rayleigh distribution and Ricean distribution tend to fail for the echoes of the first and second types of cells under the high-resolution condition. More flexible distribution forms should be selected. In this paper, the first and second types of echoes are modelled with Gamma distribution. The pdf of Gamma distribution is

$$p_{\Gamma}(x) = \begin{cases} \frac{m}{\mu \cdot \Gamma(m)} \left(\frac{mx}{\mu} \right)^{m-1} \exp\left(-\frac{mx}{\mu}\right), & x > 0 \\ 0, & x \leq 0 \end{cases} \quad (2)$$

where μ is the mean, m is the order parameter, and $\Gamma(\cdot)$ denotes the gamma function. The reasons why Gamma distribution is selected are as follows. i) The echo value per resolution cell is larger than zero, while a Gamma variant is also larger than zero; ii) The pdfs of Gamma family are similar to those of Rayleigh distribution and Ricean distribution, all of which have “heavy tails”; iii) Gamma distribution is a flexible distribution with two parameters, i.e. mean parameter and order parameter. By changing the two parameters, Gamma family can approximate to Exponential distribution, χ^2 distribution, Gaussian distribution and so on; iv) Compared with Ricean distribution and other complex distributions, Gamma distribution is simple, of which the two parameters can be estimated by the maximum likelihood (ML) method.

b) The echo of third type of resolution cell follows a distribution with several peaks. A mixture distribution can approximate such distribution. For the sake of simplicity, this paper selects Gaussian mixture distribution.

Thus this paper proposed that echoes of different types of cells are modeled as the corresponding distribution forms and an HRRP follows the united distribution.

2.3 Distinguishing Between the Three Types of Range Resolution Cells

According to what mentioned above, as far as the composition is concerned, the numbers of predominant scatterers in the three types of cells are different; as far as the distribution form is concerned, the first and second types of echoes follow distri-

butions with one peak, but the third follows a distribution with several peaks. Therefore, the number of clusters of training data can reflect the type of a resolution cell, in other words, the first and second types of echoes should have one cluster, whereas the third should have several. Xu et al. in 1993 proposed the RPCL algorithm that can perform appropriate clustering without knowing the number of clusters by automatically driving extra seed points far away from the input data set [5]. Given a set of data $\{y_i\}_{i=1}^N$ (y_i is a C -dimensional vector) that forms J^* clusters (J^* is unknown), the RPCL algorithm is to perform clustering by learning J seed points ($J > J^*$), denoted as $\{\omega_j^{(0)}\}_{j=1}^J$. RPCL algorithm mainly consists of the following two steps.

Let $t = 0$, $i = 1$ and Th be a small positive number.

Step 1: Let

$$I(j|y_i) = \begin{cases} 1 & , \quad \text{if } j = c \\ -1 & , \quad \text{if } j = r \\ 0 & , \quad \text{otherwise} \end{cases} \quad 1 \leq j \leq J . \quad (3)$$

with

$$c = \arg \min_j \gamma_j \|y_i - \omega_j\|^2, \quad r = \arg \min_{j \neq c} \gamma_j \|y_i - \omega_j\|^2 . \quad (4)$$

where $\gamma_j = \frac{n_j}{\sum_{r=1}^J n_r}$ is the relative winning frequency of the seed point ω_j in the past, and n_j is the cumulative number of occurrences of $I(j|y_i) = 1$ in the past;

Step 2: Update the weight vector $\{\omega_j^{(i)}\}_{j=1}^J$ by

$$\omega_j^{(i)} |^{new} = \omega_j^{(i)} |^{old} + \Delta \omega_j . \quad (5)$$

with

$$\Delta \omega_j = \begin{cases} \alpha_c (y_i - \omega_j^{(i)}) & , \quad \text{if } I(j|y_i) = 1 \\ -\alpha_r (y_i - \omega_j^{(i)}) & , \quad \text{if } I(j|y_i) = -1 \\ 0 & , \quad \text{otherwise} \end{cases} \quad 1 \leq j \leq J . \quad (6)$$

where $0 \leq \alpha_c, \alpha_r \leq 1$ are the learning rates for the winner and rival unit, and often set $\alpha_r \square \alpha_c$ as shown in the literature [5]. Let $i = i + 1$. If $i \leq N$, the algorithm returns to Step 1.

If $i > N$ and $\|\Delta \omega_1^T, \Delta \omega_2^T, \dots, \Delta \omega_J^T\| > Th$, let $t = t + 1$ and $i = 0$. Then the algorithm returns to Step 1. If $i > N$ and $\|\Delta \omega_1^T, \Delta \omega_2^T, \dots, \Delta \omega_J^T\| \leq Th$, the iteration ends. According to the number of samples in each cluster, the cluster label is defined as

$$Cluster_Index(j) = \begin{cases} 1 & , \quad \text{if } Num\{I(j|y_i) = 1 | i = 1, 2, \dots, N\} > 0 \\ 0 & , \quad \text{if } Num\{I(j|y_i) = 1 | i = 1, 2, \dots, N\} = 0 \end{cases} \quad 1 \leq j \leq J . \quad (7)$$

Then the number of clusters after the competitive learning is

$$N_c = Num\{Cluster_Index(j) = 1 | j = 1, 2, \dots, J\} . \quad (8)$$

When the number of clusters per resolution cell is estimated, its type can be distinguished and its echo can be modeled as the corresponding distribution form. Thus we can build a compound statistical model to represent the statistical characteristics of target HRRP. Parameter estimation of Gamma distribution and Gaussian mixture distribution is achieved by using the ML method [8] and the expectation-maximization (EM) algorithm [7], respectively.

3 Recognition Experiments Based on Measured Data

The recognition experiments performed here are based on real airplane data. Fig. 1 shows the approximated pdfs of the echoes of some typical resolution cells, where the bar represents that approximated by histogram method, the dotted line represents that by Pazern windows using Gaussian kernels and the real line represents that by the statistical model proposed in this paper. There are two mixture components in the third type of cells in our experiment. The proportions of the third type of cells are 28.00%, 30.50% and 47.00% for “Yark-42”, “Cessna” and “An-26”, respectively. Table 1 shows a comparison in recognition performance between Gaussian model with the optimal power transformation ($\alpha^*=0.5$), Gamma model and the proposed compound model. Obviously, our model has the largest average recognition rate. Moreover, in order to compare the effect of noises on the three models, Gaussian white noises with their powers determined by the SNR are added to the I- and Q-echo of each resolution cell, respectively. The average recognition rates of the three models versus the SNR are shown in Fig. 2. Obviously, Gaussian model with power transformation is the most sensitive to noises. With SNR=20 dB, the average recognition rates of Gaussian model, Gamma model and our compound model are 60.67%, 70.67% and 80.13%, respectively.

Table 1. Confusion matrices of Gaussian model, Gamma model and the compound model

	Gaussian model $\alpha^*=0.5$			Gamma model			compound model		
	Yark-42	Cessna	An-26	Yark-42	Cessna	An-26	Yark-42	Cessna	An-26
Yark-42	99.50	1.05	16.05	99.25	0.30	12.25	100.00	0.80	6.40
Cessna	0	96.00	9.95	0.25	91.65	4.40	0	94.40	0.80
An-26	0.50	2.95	74.00	0.50	8.05	83.35	0	4.80	92.80
Average		89.83			91.42			95.73	

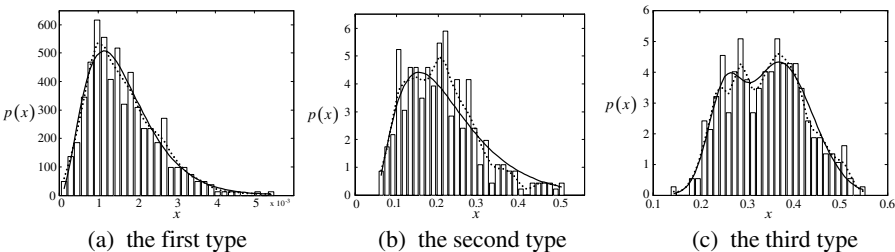


Fig. 1. the approximated pdfs of the echoes of some typical resolution cells

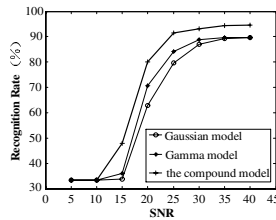


Fig. 2. The average recognition rates versus the SNR

4 Conclusions

There are two statistical models available for target HRRP, i.e. Gaussian model and Gamma model. This paper develops a compound statistical model comprising two distribution forms, i.e. Gamma distribution and Gaussian mixture distribution, by which echoes of different types of resolution cells are modeled as the corresponding distribution forms and an HRRP follows the united distribution. Determination of the type of a resolution cell is achieved by using the RPCL algorithm. Experimental results for measured data show that the proposed model not only has better recognition performance but also is more robust to noises than Gaussian model and Gamma model.

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